

Significant Figures

1. Rules for Counting Significant Figures (*sf* or *sig. figs*)

- All non-zero digits are significant.
(e.g. 123 has 3 sf)
- Zeros between non-zero digits are significant.
(e.g. 12304 has 5 sf)
- Leading zeros are not significant.
(e.g. 0.0012 has 2 sf)
- Trailing zeros are significant only if a decimal point is present.
(e.g. 120 has 2 sf; 120. has 3 sf; 120.30 has 5 sf)

2. Addition & Subtraction

- Round the result to the same *decimal place* as the **least precise** term.
i.e. count the number of digits after the decimal place for each addend, subtrahend, and minuend.
Keep the minimum number of digits after the decimal place in your answer.

$$12.11 + 0.3 + 1.025 = 13.435 \xrightarrow{\text{round to tenths}} 13.4$$

3. Multiplication & Division

- Round the result to the same *total significant figures* as the factor with the fewest.

$$4.56 \times 1.4 = 6.384 \xrightarrow{\text{round to 2 sf}} 6.4$$

Calculations involving Measured Values

Keep an infinite number of significant figures during intermediate steps in calculation. Keep track of the proper number of significant figures and only round the final answer.

Step Explanation	Calculation
1. Given measurements: • Mass of a Beaker + Sample = 125.43 g (5 sf) • Mass of a Beaker = 123.68 g (5 sf) • Volume = 13.5 mL (3 sf) • Find the mass of sample in 39.0 mL (3 sf)	
2. Calculations (infinite precision): Retain every digit in your calculator:	$\frac{125.43 \text{ g} - 123.68 \text{ g}}{13.5 \text{ mL}} = \frac{1.75 \text{ g}}{13.5 \text{ mL}} = 0.12963 \dots \text{ g/mL}$
3. Applying Significant-Figure Rules: Answer must have 3 sf (same as least-precise input).	$(0.12962963 \dots \text{ g/mL}) 39.0 \text{ mL} = 5.0555555 \dots \text{ g}$
4. Final rounded result: Round the infinite-precision quotient to 3 sf:	$5.0555555 \dots \text{ g} \xrightarrow{\text{round to 3 sf}} \text{5.06 g} \xleftarrow{\text{correct!}}$

Note: if rounding to 0.130 g/mL in step 2, the wrong answer would be found. Do not round early.

$$(0.130 \text{ g/mL}) 39.0 \text{ mL} = \text{5.07 g} \xleftarrow{\text{wrong!}}$$